

FagusED · A-Level 9700
1 Cell structure

What is the central idea in Cell structure?

9700-01-C1

QUESTION

FagusED · A-Level 9700
1 Cell structure

Which mechanism, condition or distinction matters in Cell structure?

9700-01-C2

QUESTION

FagusED · A-Level 9700
1 Cell structure

What determines the smallest resolvable detail in a micrograph?
A Resolution
B Image colour
C Printed page size
D Number of labels

9700-01-M

QUESTION

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1 Cell structure

A micrograph shows a 12 mm chloroplast that is actually 6 μm long. Calculate magnification and name one organelle present in plant cells but absent from typical animal cells.

9700-01-P

QUESTION

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2 Biological molecules

What is the central idea in Biological molecules?

9700-02-C1

QUESTION

FagusED · A-Level 9700
2 Biological molecules

Which mechanism, condition or distinction matters in Biological molecules?

9700-02-C2

QUESTION

FagusED · A-Level 9700
2 Biological molecules

What bond joins amino acids in a protein?
A Glycosidic
B Peptide
C Ester
D Hydrogen only

9700-02-M

QUESTION

FagusED · A-Level 9700
2 Biological molecules

How many water molecules form when one chain of seven amino acids assembles from free amino acids? Explain.

9700-02-P

QUESTION

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3 Enzymes

What is the central idea in Enzymes?

9700-03-C1

QUESTION

FagusED · A-Level 9700
3 Enzymes

Which mechanism, condition or distinction matters in Enzymes?

9700-03-C2

QUESTION

FagusED · A-Level 9700
3 Enzymes

What happens to a competitive inhibitor's effect when substrate increases strongly?
A It always becomes stronger
B The enzyme changes into product
C It may be overcome
D Temperature becomes irrelevant

9700-03-M

QUESTION

FagusED · A-Level 9700
3 Enzymes

Two enzymes have equal maximum activity, but enzyme X needs higher substrate concentration to reach half-maximal rate. Give one possible explanation and how to test it.

9700-03-P

QUESTION

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1 Cell structure

A. Resolution
Why: Light microscopes resolve cells and tissues; electron microscopes resolve organelles and membranes.

9700-01-M

ANSWER

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1 Cell structure

Prokaryotes lack a membrane-bound nucleus and 80S ribosomes; eukaryotic organelles compartmentalise metabolic reactions. Resolution, unlike magnification, determines the smallest separable detail.

9700-01-C2

ANSWER

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1 Cell structure

Light microscopes resolve cells and tissues; electron microscopes resolve organelles and membranes. Magnification = image size/actual size, with matching units.

9700-01-C1

ANSWER

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2 Biological molecules

The sequence of amino acids determines how a protein folds and thus its specificity; denaturation disrupts higher structure without usually hydrolysing peptide bonds.

9700-02-C2

ANSWER

FagusED · A-Level 9700
2 Biological molecules

Monomers condense into polymers, releasing water; hydrolysis breaks bonds with water. Carbohydrates have glycosidic bonds, proteins peptide bonds and nucleic acids phosphodiester bonds.

9700-02-C1

ANSWER

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1 Cell structure

12 mm = 12 000 μm ; magnification = 2000. Chloroplast or large permanent vacuole.

9700-01-P

ANSWER

FagusED · A-Level 9700
3 Enzymes

Enzymes lower activation energy by binding substrate at an active site and stabilising a transition state. Initial rate is a fair comparison because substrate depletion and product accumulation are smallest early on.

9700-03-C1

ANSWER

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2 Biological molecules

Six peptide bonds are formed, so six condensation events produce six water molecules.

9700-02-P

ANSWER

FagusED · A-Level 9700
2 Biological molecules

B. Peptide
Why: Monomers condense into polymers, releasing water; hydrolysis breaks bonds with water.

9700-02-M

ANSWER

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3 Enzymes

Competitive inhibition of X is possible; repeat with increasing substrate concentrations; if inhibition can be overcome at high substrate, both approach similar maximum rate.

9700-03-P

ANSWER

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3 Enzymes

C. It may be overcome
Why: Enzymes lower activation energy by binding substrate at an active site and stabilising a transition state.

9700-03-M

ANSWER

FagusED · A-Level 9700
3 Enzymes

Temperature increases collision rates until heat disrupts protein structure; pH can change ionisation of active-site groups. Competitive inhibitors increase apparent substrate needed; non-competitive inhibitors lower maximum rate.

9700-03-C2

ANSWER

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4 Cell membranes and transport

What is the central idea in Cell membranes and transport?

9700-04-C1

QUESTION

FagusED · A-Level 9700
4 Cell membranes and transport

Which mechanism, condition or distinction matters in Cell membranes and transport?

9700-04-C2

QUESTION

FagusED · A-Level 9700
4 Cell membranes and transport

What directly drives active transport against a gradient?
A Osmosis
B Simple diffusion
C Gravity
D ATP use

9700-04-M

QUESTION

FagusED · A-Level 9700
4 Cell membranes and transport

Explain why a plant cell loses turgor in concentrated sucrose but may stay viable.

9700-04-P

QUESTION

FagusED · A-Level 9700
5 The mitotic cell cycle

What is the central idea in The mitotic cell cycle?

9700-05-C1

QUESTION

FagusED · A-Level 9700
5 The mitotic cell cycle

Which mechanism, condition or distinction matters in The mitotic cell cycle?

9700-05-C2

QUESTION

FagusED · A-Level 9700
5 The mitotic cell cycle

What separates to opposite poles during mitosis?
A Sister chromatids
B Homologous chromosomes only
C Cell walls
D Two intact nuclei

9700-05-M

QUESTION

FagusED · A-Level 9700
5 The mitotic cell cycle

A field has 18 dividing cells among 150 cells. Calculate mitotic index and explain why daughter cells have equal chromosome number.

9700-05-P

QUESTION

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6 Nucleic acids and protein synthesis

What is the central idea in Nucleic acids and protein synthesis?

9700-06-C1

QUESTION

FagusED · A-Level 9700
6 Nucleic acids and protein synthesis

Which mechanism, condition or distinction matters in Nucleic acids and protein synthesis?

9700-06-C2

QUESTION

FagusED · A-Level 9700
6 Nucleic acids and protein synthesis

Which molecule carries codons read by ribosomes?
A DNA polymerase
B mRNA
C ATP synthase
D Haemoglobin

9700-06-M

QUESTION

FagusED · A-Level 9700
6 Nucleic acids and protein synthesis

A DNA template segment 3'-TAC GAA-5' is transcribed. Give mRNA 5'-3' and explain why changing one base does not always change protein.

9700-06-P

QUESTION

FagusED · A-Level 9700
7 Transport in plants

What is the central idea in Transport in plants?

9700-07-C1

QUESTION

FagusED · A-Level 9700
7 Transport in plants

Which mechanism, condition or distinction matters in Transport in plants?

9700-07-C2

QUESTION

FagusED · A-Level 9700
7 Transport in plants

Which tissue translocates sucrose from source to sink?
A Xylem vessel
B Epidermis
C Phloem sieve tubes
D Palisade cells

9700-07-M

QUESTION

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4 Cell membranes and transport

D. ATP use
Why: A phospholipid bilayer has hydrophilic heads facing water and hydrophobic tails inward.

9700-04-M

ANSWER

FagusED · A-Level 9700
4 Cell membranes and transport

Diffusion proceeds down a concentration gradient; osmosis is net water movement from higher to lower water potential through a partially permeable membrane. Active transport uses ATP to move substances against gradients.

9700-04-C2

ANSWER

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4 Cell membranes and transport

A phospholipid bilayer has hydrophilic heads facing water and hydrophobic tails inward. Proteins support facilitated diffusion, active transport and signalling.

9700-04-C1

ANSWER

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5 The mitotic cell cycle

Cell cycle checkpoints restrict division after DNA damage; uncontrolled proliferation can lead to tumours. A mitotic index estimates the fraction of observed cells undergoing mitosis.

9700-05-C2

ANSWER

FagusED · A-Level 9700
5 The mitotic cell cycle

Interphase includes DNA replication before mitosis. Mitosis separates identical sister chromatids so daughter nuclei retain chromosome number and genetic identity, subject to mutation.

9700-05-C1

ANSWER

FagusED · A-Level 9700
4 Cell membranes and transport

Outside solution has lower water potential; water exits by osmosis through the partially permeable membrane; membrane pulls from wall (plasmolysis), but wall remains intact and cell can recover.

9700-04-P

ANSWER

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6 Nucleic acids and protein synthesis

DNA is antiparallel, with A-T and C-G complementary pairing. Semi-conservative replication yields one old and one newly synthesised strand per molecule.

9700-06-C1

ANSWER

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5 The mitotic cell cycle

Index = $18/150 = 0.12$. DNA replicated in interphase, then sister chromatids separate to opposite poles.

9700-05-P

ANSWER

FagusED · A-Level 9700
5 The mitotic cell cycle

A. Sister chromatids
Why: Interphase includes DNA replication before mitosis.

9700-05-M

ANSWER

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6 Nucleic acids and protein synthesis

mRNA 5'-AUG CUU-3'. Genetic code is degenerate: a changed codon can specify the same amino acid.

9700-06-P

ANSWER

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6 Nucleic acids and protein synthesis

B. mRNA
Why: DNA is antiparallel, with A-T and C-G complementary pairing.

9700-06-M

ANSWER

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6 Nucleic acids and protein synthesis

Transcription produces RNA from a DNA template; translation reads mRNA codons at ribosomes while tRNA brings amino acids. A substitution may be silent, missense or nonsense depending on codon and reading frame.

9700-06-C2

ANSWER

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7 Transport in plants

C. Phloem sieve tubes
Why: Xylem dead lignified vessels carry water upward under transpiration pull; cohesion and adhesion maintain water columns.

9700-07-M

ANSWER

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7 Transport in plants

Phloem living sieve tubes translocate sucrose from sources to sinks, driven by loading, osmotic water movement and pressure flow. Water potential differences at root, leaf and atmosphere sustain net transport.

9700-07-C2

ANSWER

FagusED · A-Level 9700
7 Transport in plants

Xylem dead lignified vessels carry water upward under transpiration pull; cohesion and adhesion maintain water columns.

9700-07-C1

ANSWER

FagusED · A-Level 9700
7 Transport in plants

Describe how increasing wind speed may change transpiration rate and why the effect can plateau.

9700-07-P

QUESTION

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8 Transport in mammals

What is the central idea in Transport in mammals?

9700-08-C1

QUESTION

FagusED · A-Level 9700
8 Transport in mammals

Which mechanism, condition or distinction matters in Transport in mammals?

9700-08-C2

QUESTION

FagusED · A-Level 9700
8 Transport in mammals

What is the Bohr effect useful for in active tissue?
A More blood clotting
B Lower CO₂
C No diffusion
D Oxygen unloading

9700-08-M

QUESTION

FagusED · A-Level 9700
8 Transport in mammals

Explain two structural adaptations of a capillary for exchange and why oxygen unloading is greater in active tissues.

9700-08-P

QUESTION

FagusED · A-Level 9700
9 Gas exchange

What is the central idea in Gas exchange?

9700-09-C1

QUESTION

FagusED · A-Level 9700
9 Gas exchange

Which mechanism, condition or distinction matters in Gas exchange?

9700-09-C2

QUESTION

FagusED · A-Level 9700
9 Gas exchange

What adaptation lowers diffusion distance at alveoli?
A Thin epithelium
B Thick cartilage
C Small total area
D No blood supply

9700-09-M

QUESTION

FagusED · A-Level 9700
9 Gas exchange

Explain why pulmonary fibrosis reduces oxygen transfer even if alveolar oxygen partial pressure stays constant.

9700-09-P

QUESTION

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10 Infectious diseases

What is the central idea in Infectious diseases?

9700-10-C1

QUESTION

FagusED · A-Level 9700
10 Infectious diseases

Which mechanism, condition or distinction matters in Infectious diseases?

9700-10-C2

QUESTION

FagusED · A-Level 9700
10 Infectious diseases

Why do antibiotics not directly treat viral replication?
A Viruses have more ribosomes
B Viruses lack targeted bacterial processes
C Antibiotics only work in air
D All viruses are bacteria

9700-10-M

QUESTION

FagusED · A-Level 9700
8 Transport in mammals

Haemoglobin's cooperative oxygen binding produces a sigmoidal dissociation curve; a lower affinity in metabolically active tissue assists unloading.

9700-08-C2

ANSWER

FagusED · A-Level 9700
8 Transport in mammals

Double circulation sends blood through pulmonary and systemic circuits; separation permits high systemic pressure and lower pulmonary pressure. Arteries have thick elastic walls, veins valves and larger lumens; capillaries have thin walls for exchange.

9700-08-C1

ANSWER

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7 Transport in plants

Wind removes humid boundary layer, increasing water-vapour gradient and evaporation; rate may become limited by stomatal aperture/water supply or closure in water stress.

9700-07-P

ANSWER

FagusED · A-Level 9700
9 Gas exchange

Ventilation refreshes alveolar air and maintains oxygen and carbon dioxide partial-pressure gradients. Numerous alveoli provide a large surface, a thin epithelium short diffusion path and capillary networks carry gases away.

9700-09-C1

ANSWER

FagusED · A-Level 9700
8 Transport in mammals

Thin endothelial wall minimises diffusion distance; narrow lumen/slower flow or extensive network increases exchange time/surface; high CO₂ lowers haemoglobin affinity (Bohr effect).

9700-08-P

ANSWER

FagusED · A-Level 9700
8 Transport in mammals

D. Oxygen unloading
Why: Double circulation sends blood through pulmonary and systemic circuits; separation permits high systemic pressure and lower pulmonary pressure.

9700-08-M

ANSWER

FagusED · A-Level 9700
9 Gas exchange

Fibrosis thickens diffusion barrier; diffusion distance increases, slowing oxygen uptake into blood.

9700-09-P

ANSWER

FagusED · A-Level 9700
9 Gas exchange

A. Thin epithelium
Why: Ventilation refreshes alveolar air and maintains oxygen and carbon dioxide partial-pressure gradients.

9700-09-M

ANSWER

FagusED · A-Level 9700
9 Gas exchange

Insects have tracheae and tracheoles delivering air toward cells, while stomata regulate plant gas exchange.

9700-09-C2

ANSWER

FagusED · A-Level 9700
10 Infectious diseases

B. Viruses lack targeted bacterial processes
Why: A pathogen causes transmissible disease; viruses replicate inside host cells, while bacteria may divide independently.

9700-10-M

ANSWER

FagusED · A-Level 9700
10 Infectious diseases

Antibiotics act on bacterial structures/processes but do not treat viruses. Resistance evolves when a heritable variant survives exposure and reproduces.

9700-10-C2

ANSWER

FagusED · A-Level 9700
10 Infectious diseases

A pathogen causes transmissible disease; viruses replicate inside host cells, while bacteria may divide independently. Transmission can involve droplets, water or vectors, so prevention must target the route.

9700-10-C1

ANSWER

FagusED · A-Level 9700
10 Infectious diseases

A bacterial population survives an antibiotic treatment. Explain selection of resistance in three linked steps.

9700-10-P

QUESTION

FagusED · A-Level 9700
11 Immunity

What is the central idea in Immunity?

9700-11-C1

QUESTION

FagusED · A-Level 9700
11 Immunity

Which mechanism, condition or distinction matters in Immunity?

9700-11-C2

QUESTION

FagusED · A-Level 9700
11 Immunity

What enables a faster secondary antibody response?
A Erythrocytes
B Platelets
C Memory cells
D Antigens disappear forever

9700-11-M

QUESTION

FagusED · A-Level 9700
11 Immunity

Distinguish active vaccination from passive antibody transfer in onset and duration of protection.

9700-11-P

QUESTION

FagusED · A-Level 9700
12 Energy and respiration

What is the central idea in Energy and respiration?

9700-12-C1

QUESTION

FagusED · A-Level 9700
12 Energy and respiration

Which mechanism, condition or distinction matters in Energy and respiration?

9700-12-C2

QUESTION

FagusED · A-Level 9700
12 Energy and respiration

What is oxygen's role in aerobic respiration?
A First glycolysis substrate
B Only carbon source
C ATP synthase motor
D Final electron acceptor

9700-12-M

QUESTION

FagusED · A-Level 9700
12 Energy and respiration

Explain why cyanide blocking electron transport rapidly lowers ATP production despite available glucose.

9700-12-P

QUESTION

FagusED · A-Level 9700
13 Photosynthesis

What is the central idea in Photosynthesis?

9700-13-C1

QUESTION

FagusED · A-Level 9700
13 Photosynthesis

Which mechanism, condition or distinction matters in Photosynthesis?

9700-13-C2

QUESTION

FagusED · A-Level 9700
13 Photosynthesis

Where does Calvin cycle carbon fixation occur?
A Stroma
B Thylakoid lumen
C Mitochondrial matrix
D Cytoplasm only

9700-13-M

QUESTION

FagusED · A-Level 9700
11 Immunity

T helper cells coordinate responses; T killer cells destroy infected cells. A vaccine supplies antigen safely to induce immune memory; it does not directly deliver permanent ready-made antibodies.

9700-11-C2

ANSWER

FagusED · A-Level 9700
11 Immunity

Phagocytes engulf pathogens non-specifically; lymphocytes respond to specific antigens. B cells differentiate into plasma cells producing antibodies and memory cells enabling faster secondary responses.

9700-11-C1

ANSWER

FagusED · A-Level 9700
10 Infectious diseases

Heritable resistance variants pre-exist or arise by mutation; antibiotic kills susceptible cells so resistant cells survive; resistant cells reproduce and allele frequency rises.

9700-10-P

ANSWER

FagusED · A-Level 9700
12 Energy and respiration

Glycolysis in cytoplasm converts glucose to pyruvate with a small ATP yield; link reaction and Krebs cycle feed reduced coenzymes into electron transport.

9700-12-C1

ANSWER

FagusED · A-Level 9700
11 Immunity

Passive transfer supplies antibodies and acts immediately, but is temporary as antibodies degrade; active vaccination induces memory cells and lasts longer after a slower primary response.

9700-11-P

ANSWER

FagusED · A-Level 9700
11 Immunity

C. Memory cells
Why: Phagocytes engulf pathogens non-specifically; lymphocytes respond to specific antigens.

9700-11-M

ANSWER

FagusED · A-Level 9700
12 Energy and respiration

Electrons cannot flow to oxygen; proton gradient is not maintained, so ATP synthase makes far less ATP; only glycolysis remains.

9700-12-P

ANSWER

FagusED · A-Level 9700
12 Energy and respiration

D. Final electron acceptor
Why: Glycolysis in cytoplasm converts glucose to pyruvate with a small ATP yield; link reaction and Krebs cycle feed reduced coenzymes into electron transport.

9700-12-M

ANSWER

FagusED · A-Level 9700
12 Energy and respiration

Oxidative phosphorylation uses a proton gradient across inner mitochondrial membrane to drive ATP synthase. Oxygen is final electron acceptor. Anaerobic pathways regenerate NAD so glycolysis can continue.

9700-12-C2

ANSWER

FagusED · A-Level 9700
13 Photosynthesis

A. Stroma
Why: Photosystems in thylakoids absorb light, split water and make ATP and reduced NADP.

9700-13-M

ANSWER

FagusED · A-Level 9700
13 Photosynthesis

Light intensity, CO₂ and temperature may each limit rate, with changing limiting factor as conditions change.

9700-13-C2

ANSWER

FagusED · A-Level 9700
13 Photosynthesis

Photosystems in thylakoids absorb light, split water and make ATP and reduced NADP. The Calvin cycle in the stroma uses ATP and reduced NADP to fix CO₂ into triose phosphate; RuBP is regenerated.

9700-13-C1

ANSWER

FagusED · A-Level 9700
13 Photosynthesis

A plant's oxygen production plateaus when light intensity doubles. State two possible limits and suggest one controlled test.

9700-13-P

QUESTION

FagusED · A-Level 9700
14 Homeostasis

What is the central idea in Homeostasis?

9700-14-C1

QUESTION

FagusED · A-Level 9700
14 Homeostasis

Which mechanism, condition or distinction matters in Homeostasis?

9700-14-C2

QUESTION

FagusED · A-Level 9700
14 Homeostasis

Which hormone usually increases after blood glucose rises?
A Glucagon
B Insulin
C ADH always
D Adrenaline only

9700-14-M

QUESTION

FagusED · A-Level 9700
14 Homeostasis

Explain the response of insulin and glucagon after a carbohydrate-rich meal.

9700-14-P

QUESTION

FagusED · A-Level 9700
15 Control and coordination

What is the central idea in Control and coordination?

9700-15-C1

QUESTION

FagusED · A-Level 9700
15 Control and coordination

Which mechanism, condition or distinction matters in Control and coordination?

9700-15-C2

QUESTION

FagusED · A-Level 9700
15 Control and coordination

Why can myelin increase impulse conduction speed?
A It removes ion channels everywhere
B It reverses impulse direction
C Impulses regenerate at nodes
D It makes synapses electrical

9700-15-M

QUESTION

FagusED · A-Level 9700
15 Control and coordination

Explain how myelination increases conduction speed and why chemical synapse transmission is directional.

9700-15-P

QUESTION

FagusED · A-Level 9700
16 Inheritance

What is the central idea in Inheritance?

9700-16-C1

QUESTION

FagusED · A-Level 9700
16 Inheritance

Which mechanism, condition or distinction matters in Inheritance?

9700-16-C2

QUESTION

FagusED · A-Level 9700
16 Inheritance

What can crossing over during meiosis produce?
A Identical clones only
B Doubled chromosome number always
C Loss of all genes
D New allele combinations

9700-16-M

QUESTION

FagusED · A-Level 9700
14 Homeostasis

Kidney nephrons filter blood and selectively reabsorb useful solutes; ADH increases water permeability of collecting ducts.

9700-14-C2

ANSWER

FagusED · A-Level 9700
14 Homeostasis

Negative feedback detects deviation from a set point and produces a response opposing the change. Pancreatic β cells release insulin at high blood glucose, increasing uptake and glycogen synthesis; α cells release glucagon at low glucose, promoting glycogen breakdown and gluconeogenesis.

9700-14-C1

ANSWER

FagusED · A-Level 9700
13 Photosynthesis

CO₂ availability or enzyme temperature may limit; vary CO₂ while holding light and temperature constant and observe whether oxygen production rises.

9700-13-P

ANSWER

FagusED · A-Level 9700
15 Control and coordination

A nervous impulse follows depolarisation when voltage-gated sodium channels open and repolarisation via potassium channels. Myelin permits saltatory conduction.

9700-15-C1

ANSWER

FagusED · A-Level 9700
14 Homeostasis

Blood glucose increases; insulin increases and promotes cellular uptake/storage as glycogen; glucagon secretion decreases.

9700-14-P

ANSWER

FagusED · A-Level 9700
14 Homeostasis

B. Insulin
Why: Negative feedback detects deviation from a set point and produces a response opposing the change.

9700-14-M

ANSWER

FagusED · A-Level 9700
15 Control and coordination

Action potentials regenerate at nodes of Ranvier, reducing membrane to depolarise; transmitter vesicles are presynaptic and receptor proteins postsynaptic.

9700-15-P

ANSWER

FagusED · A-Level 9700
15 Control and coordination

C. Impulses regenerate at nodes
Why: A nervous impulse follows depolarisation when voltage-gated sodium channels open and repolarisation via potassium channels.

9700-15-M

ANSWER

FagusED · A-Level 9700
15 Control and coordination

Chemical synapses release neurotransmitter following calcium influx; transmission is one way because vesicles and receptor sites differ across the cleft. Hormones travel in blood and generally act more slowly but for longer.

9700-15-C2

ANSWER

FagusED · A-Level 9700
16 Inheritance

D. New allele combinations
Why: Meiosis produces haploid gametes and variation by crossing over and independent assortment.

9700-16-M

ANSWER

FagusED · A-Level 9700
16 Inheritance

For independently assorting genes, multiply probabilities; for linkage, recombinant classes are less frequent than predicted by independent assortment. A chi-square test compares observed and expected counts but does not prove the null hypothesis true.

9700-16-C2

ANSWER

FagusED · A-Level 9700
16 Inheritance

Meiosis produces haploid gametes and variation by crossing over and independent assortment. At fertilisation diploid number is restored.

9700-16-C1

ANSWER

FagusED · A-Level 9700
16 Inheritance

Two heterozygous parents produce 80 offspring; 18 show a recessive phenotype. Calculate expected number under a 3:1 ratio and give one cause of deviation.

9700-16-P

QUESTION

FagusED · A-Level 9700
17 Selection and evolution

What is the central idea in Selection and evolution?

9700-17-C1

QUESTION

FagusED · A-Level 9700
17 Selection and evolution

Which mechanism, condition or distinction matters in Selection and evolution?

9700-17-C2

QUESTION

FagusED · A-Level 9700
17 Selection and evolution

At the population level, what changes during evolution?
A Allele frequencies
B Each organism's inherited DNA by need
C Species names only
D Every trait equally

9700-17-M

QUESTION

FagusED · A-Level 9700
17 Selection and evolution

In a population exposed to a new drug, explain why resistance increases over generations without claiming individual organisms adapt by need.

9700-17-P

QUESTION

FagusED · A-Level 9700
18 Classification, biodiversity and conserv...

What is the central idea in Classification, biodiversity and conservation?

9700-18-C1

QUESTION

FagusED · A-Level 9700
18 Classification, biodiversity and conserv...

Which mechanism, condition or distinction matters in Classification, biodiversity and conservation?

9700-18-C2

QUESTION

FagusED · A-Level 9700
18 Classification, biodiversity and conserv...

Which pair matters for species diversity?
A Climate and pH only
B Richness and evenness
C Only total individual count
D Only DNA molecule length

9700-18-M

QUESTION

FagusED · A-Level 9700
18 Classification, biodiversity and conserv...

State two reasons genetic diversity matters to a threatened population and one limitation of ex situ conservation.

9700-18-P

QUESTION

FagusED · A-Level 9700
19 Genetic technology

What is the central idea in Genetic technology?

9700-19-C1

QUESTION

FagusED · A-Level 9700
19 Genetic technology

Which mechanism, condition or distinction matters in Genetic technology?

9700-19-C2

QUESTION

FagusED · A-Level 9700
19 Genetic technology

Which enzyme extends primers in PCR?
A DNA ligase
B Restriction endonuclease
C Thermostable DNA polymerase
D RNA polymerase only

9700-19-M

QUESTION

FagusED · A-Level 9700
17 Selection and evolution

Genetic drift can cause chance changes, especially in small populations. Speciation requires reproductive isolation followed by genetic divergence. Selection acts on phenotypes, while populations evolve across generations.

9700-17-C2

ANSWER

FagusED · A-Level 9700
17 Selection and evolution

Mutation produces new alleles; natural selection changes allele frequencies when variants affect survival and reproductive success in a given environment.

9700-17-C1

ANSWER

FagusED · A-Level 9700
16 Inheritance

Expected recessive = $80/4 = 20$; observed 18 can differ by sampling chance; mutation/selection or flawed model are other acceptable reasons.

9700-16-P

ANSWER

FagusED · A-Level 9700
18 Classification, biodiversity and conserv...

Classification reflects evolutionary relationships using shared traits and molecular sequence data. Biodiversity can be considered at genetic, species and ecosystem levels.

9700-18-C1

ANSWER

FagusED · A-Level 9700
17 Selection and evolution

Random inherited variants arise before/during exposure; resistant individuals survive and reproduce more; resistant allele frequency rises in the population.

9700-17-P

ANSWER

FagusED · A-Level 9700
17 Selection and evolution

A. Allele frequencies
Why: Mutation produces new alleles; natural selection changes allele frequencies when variants affect survival and reproductive success in a given environment.

9700-17-M

ANSWER

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18 Classification, biodiversity and conserv...

Different alleles improve capacity to respond to environmental change and reduce inbreeding effects; ex situ populations may be small/lose habitat behaviour or require reintroduction.

9700-18-P

ANSWER

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18 Classification, biodiversity and conserv...

B. Richness and evenness
Why: Classification reflects evolutionary relationships using shared traits and molecular sequence data.

9700-18-M

ANSWER

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18 Classification, biodiversity and conserv...

A high species richness alone does not describe evenness; diversity indices combine both. Conservation can protect habitats in situ and populations ex situ; maintain genetic variation to reduce vulnerability.

9700-18-C2

ANSWER

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19 Genetic technology

C. Thermostable DNA polymerase
Why: PCR cycles denature DNA, anneal primers and extend strands with thermostable polymerase; product can double approximately each cycle under ideal conditions.

9700-19-M

ANSWER

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Restriction enzymes cut defined sequences, ligase joins DNA, and vectors introduce constructs into cells. Gel electrophoresis separates fragments by size; controls are needed to interpret band patterns and limit contamination.

9700-19-C2

ANSWER

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19 Genetic technology

PCR cycles denature DNA, anneal primers and extend strands with thermostable polymerase; product can double approximately each cycle under ideal conditions.

9700-19-C1

ANSWER

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19 Genetic technology

A PCR sample starts with 10 double-stranded targets. Estimate ideal copies after six cycles and give one reason actual yield differs.

9700-19-P

QUESTION

FagusED · A-Level 9700
P3 Advanced practical skills

What is the central idea in P3 Advanced practical skills?

9700-20-C1

QUESTION

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P3 Advanced practical skills

Which mechanism, condition or distinction matters in P3 Advanced practical skills?

9700-20-C2

QUESTION

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P3 Advanced practical skills

Why use biological replicates in a practical?
A To change several conditions
B To avoid labelling axes
C To make one measurement exact
D To estimate variation

9700-20-M

QUESTION

FagusED · A-Level 9700
P3 Advanced practical skills

A student studies sucrose concentration and potato mass change. Identify independent, dependent and two controlled variables.

9700-20-P

QUESTION

FagusED · A-Level 9700
P5 Planning, analysis and evaluation

What is the central idea in P5 Planning, analysis and evaluation?

9700-21-C1

QUESTION

FagusED · A-Level 9700
P5 Planning, analysis and evaluation

Which mechanism, condition or distinction matters in P5 Planning, analysis and evaluation?

9700-21-C2

QUESTION

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P5 Planning, analysis and evaluation

What would strengthen a proposed causal explanation?
A Controlled intervention and repeats
B A larger graph title
C One unreplicated observation
D Removing units

9700-21-M

QUESTION

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P5 Planning, analysis and evaluation

A study finds leaf area and transpiration rate correlated. Explain two reasons this does not alone establish causation and propose one improvement.

9700-21-P

QUESTION

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P3 Advanced practical skills

Use a microscope scale calibrated by stage micrometer; in enzyme or osmosis studies hold temperature, pH and initial material comparable. Present calculated rates with units and graphs with suitably labelled axes.

9700-20-C2

ANSWER

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P3 Advanced practical skills

Biology practical work requires clear variable definitions, biological replicates and consistent measurement.

9700-20-C1

ANSWER

FagusED · A-Level 9700
19 Genetic technology

$10 \times 2^6 = 640$ copies; reagent depletion, incomplete efficiency or inhibitors reduce actual yield.

9700-19-P

ANSWER

FagusED · A-Level 9700
P5 Planning, analysis and evaluation

Start with a biological hypothesis and testable prediction, then design sampling to reduce bias and enough repeats to estimate variability.

9700-21-C1

ANSWER

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P3 Advanced practical skills

Independent: sucrose concentration; dependent: change or percentage change in potato mass; controls: piece surface/starting mass, immersion time, temperature or solution volume.

9700-20-P

ANSWER

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P3 Advanced practical skills

D. To estimate variation
Why: Biology practical work requires clear variable definitions, biological replicates and consistent measurement.

9700-20-M

ANSWER

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P5 Planning, analysis and evaluation

Other factors such as stomatal density or temperature may covary; correlation gives association not controlled intervention; hold conditions constant/normalise per unit area and replicate.

9700-21-P

ANSWER

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P5 Planning, analysis and evaluation

A. Controlled intervention and repeats
Why: Start with a biological hypothesis and testable prediction, then design sampling to reduce bias and enough repeats to estimate variability.

9700-21-M

ANSWER

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P5 Planning, analysis and evaluation

Differentiate correlation from causation; choose a suitable graph/statistical test and interpret it against assumptions. Improvements should respond to limits such as uncontrolled temperature or small sample size.

9700-21-C2

ANSWER